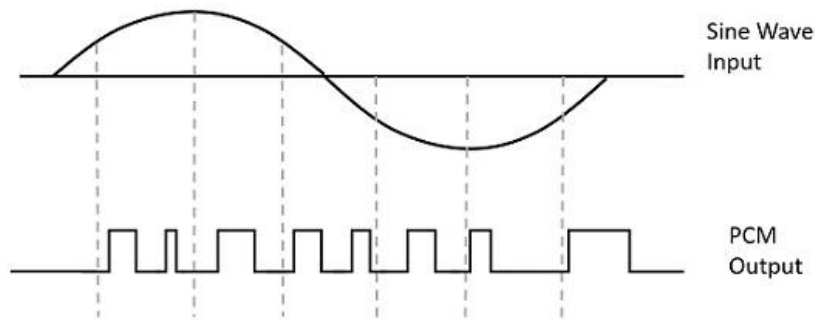


Pulse Code Modulation

A signal is Pulse Code modulated to convert its analog information into a binary sequence, i.e., 1s and 0s. The output of a **Pulse Code Modulation (PCM)** will resemble a binary sequence. The following figure shows an example of PCM output with respect to instantaneous values of a given sine wave.



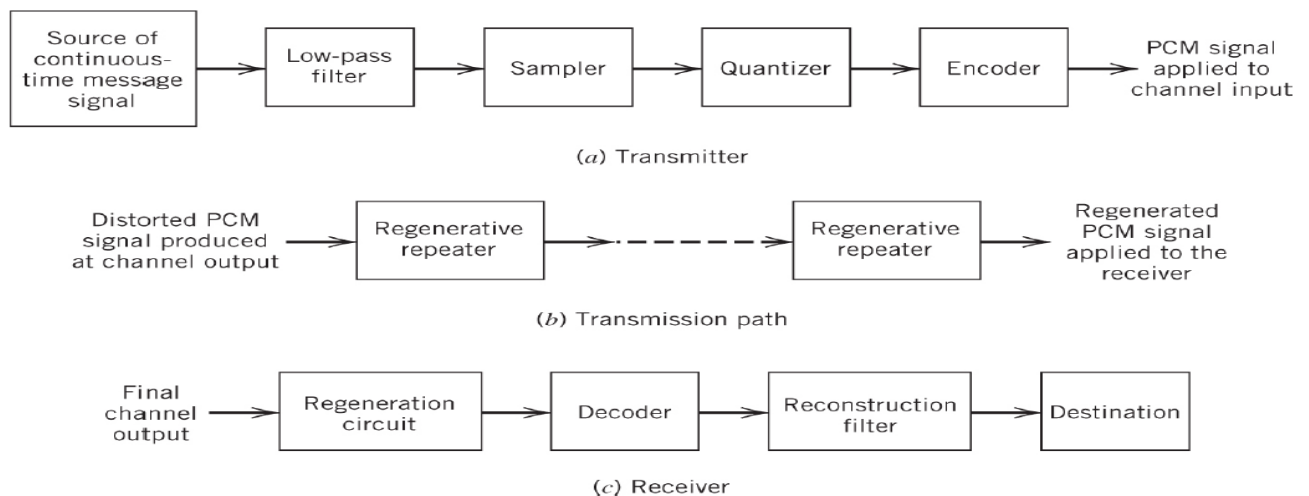
Instead of a pulse train, PCM produces a series of numbers or digits, and hence this process is called as digital. Each one of these digits, though in binary code, represent the approximate

In Pulse Code Modulation, the message signal is represented by a sequence of coded pulses. This message signal is achieved by representing the signal in discrete form in both time and amplitude of the signal sample at that instant.

Basic Elements of PCM

The transmitter section of a Pulse Code Modulator circuit consists of **Sampling, Quantizing** and **Encoding**, which are performed in the **analog-to-digital converter** section. The low pass filter prior to sampling prevents aliasing of the message signal.

The basic operations in the receiver section are **regeneration of impaired signals, decoding**, and **reconstruction** of the quantized pulse train. The following figure is the block diagram of PCM which represents the basic elements of both the transmitter and the receiver sections.



1. Low Pass Filter (LPF)

This filter eliminates the high frequency components present in the input analog signal which is greater than the highest frequency of the message signal, to avoid aliasing of the message signal.

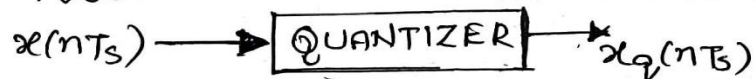
2. Sampler

This is the circuit which uses the technique that helps to collect the sample data at instantaneous values of the message signal, so as to reconstruct the original signal. The sampling rate must be greater than twice the highest frequency component W of the message signal, in accordance with the sampling theorem.

3. Quantizing (uniform quantization).

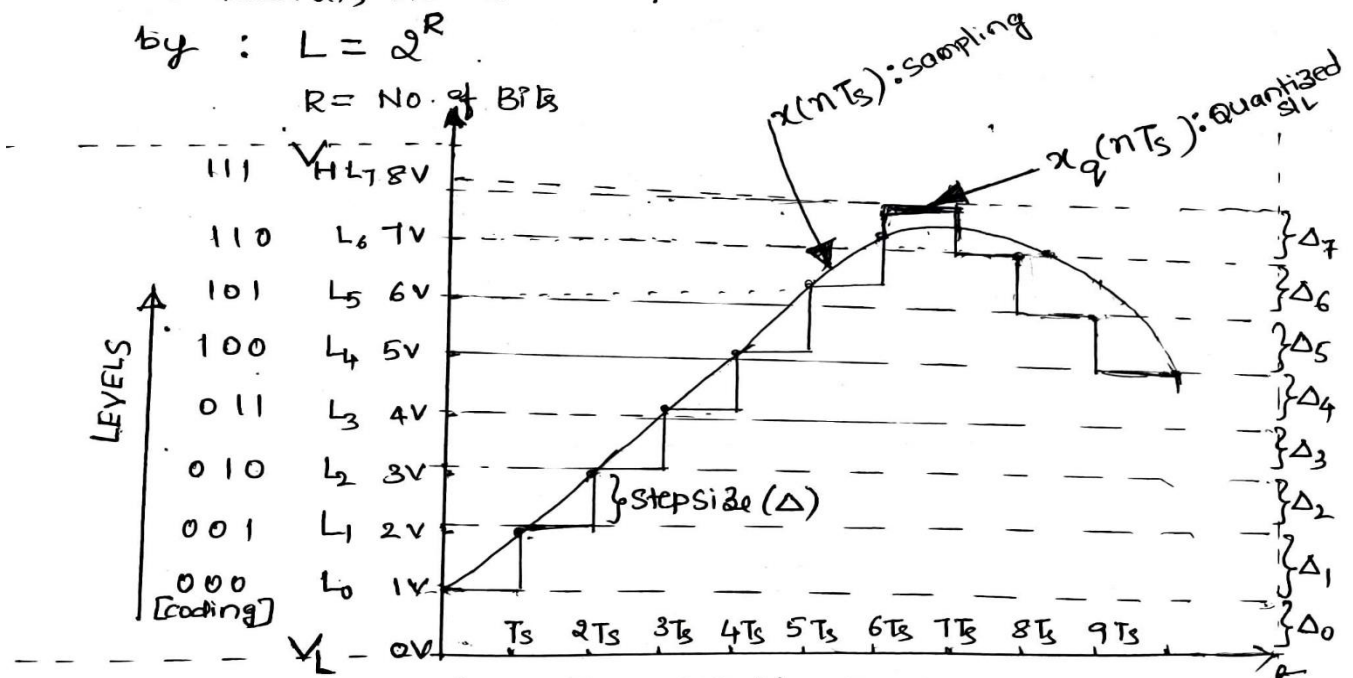
Quantizing is a process of reducing the excessive bits and confining the data. The sampled output when given to Quantizer, reduces the redundant bits and compresses the value.

Definition: The process of transforming sampled amplitude values of a message signal into a discrete amplitude values @ discrete amplitude levels is called as "QUANTIZATION".



Explanation:

- Consider a continuous time signal $x(t)$, whose excursions are confined to the voltage range V_L to V_H , as shown in Figure 5.1. Let $x(nT_s)$ be the sampled version of $x(t)$.
- Let the number of required quantizer levels is '8'. [L_0 to L_7]
- In General, the Number of levels in a quantizer is given by : $L = 2^R$



- * Step size is denoted by Δ and is given by

$$\Delta = \frac{V_H - V_L}{L}$$

where $L = 2^R$ = Number of Levels, R = Number of bits in each level

$$\therefore \text{Step Size, } \Delta = \frac{V_H - V_L}{2^R}$$

- * If the step size ' Δ ' is maintained same throughout the process of Quantization, then it is referred to as "UNIFORM QUANTIZATION".
- * The difference between the continuous amplitude level and quantized signal level is known as "Quantization error"
- i.e., Quantization error,

$$e(t) = x_q(nT_s) - x(nT_s)$$

where Quantization error varies from $+\Delta/2$ to $-\Delta/2$.

- * The random errors due to Quantization process produces a noise at the output of the quantizer and is referred to as "Quantization Noise".
- * Consider figure 5.1, the Sampling, Quantizing and coding of an analog signal is as follows.

Sampled Values of Analog signal	2V	3V	4.2V	5.2V	6.3V	7.3V	7.6V	7V
Nearest Quantizer Level	L_1	L_2	L_3	L_4	L_5	L_6	L_7	L_7
Quantizer level Voltage	2V	3V	4V	5V	6V	7V	8V	7V
Binary code	001	010	011	100	101	110	111	110

Example :

A TV-signal with a Bandwidth of 4.2 MHz is transmitted using binary PCM. The Number of representation levels is 512. calculate <i> code word length <ii> Final bit Rate.
<iii> Transmission Bandwidth. * VTU MODEL QP *

Solution :- Given : Bandwidth of TV-signal, $W = 4.2 \text{ MHz}$
Representation levels: $L = 512$, $R = ?$ [code word length]

<i> code word length : $L = 2^R$

$$\therefore \text{code word length } R = \frac{\log(L)}{\log(2)} = \frac{\log(512)}{\log(2)}$$

$$R = 9 \text{ bits}$$

<ii> Transmission Bandwidth :- Given : $W = 4.2 \times 10^6 \text{ Hz}$
 $(BW)_{\text{PCM}} = R \times f_s = 9 \times (2W) = 9 \times 2 \times 4.2 \times 10^6$
 $(BW)_{\text{PCM}} = 75.6 \text{ MHz}$

<iii> Final bit rate : Bit Rate $= R f_s = 9 \times (2W)$
 $= 9 \times 2 \times 4.2 \times 10^6 = 75.6 \text{ Mbps}$

• **Non-Uniform Quantization:**

• Problems with uniform quantization :

1. Only optimal for uniformly distributed signal .
2. Real audio signals (speech and music) are more concentrated near zeros
3. Human ear is more sensitive to quantization errors at small values

Solution -

- Using non-uniform quantization.

A non-uniform quantizer ensures smaller quantization error for small amplitude of the input signal and relatively larger step size when the input signal amplitude is large .

• **Design of Non-Uniform Quantizer:**

1. Adaptive quantizer (very complex and costly to achieve)
2. compandor with uniform quantization :

There are two popular standards for non-linear quantization known as

- (a) The μ - law companding
- (b) The A - law companding

So, why we need companding ?

Companding : is a technique that combines **compression** and **expansion**. It reduces the dynamic range of a signal (compression) before transmission and then restores it (expansion) at the receiving end. This helps to minimize noise and improve signal quality.

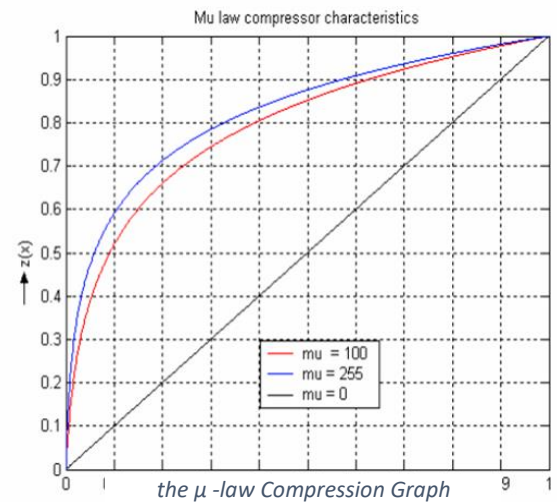
1. μ -law Companding

- **Used In:** Primarily in North America and Japan.
- **How It Works:**
 - **Compression:** The input signal is compressed using a logarithmic function. This means that smaller signals are amplified more than larger signals.

Formula: The mathematical formula for μ -law compression is:

$$F(x) = \frac{\text{sgn}(x) * \ln(1 + \mu|x|)}{\ln(1 + \mu)} \quad 0 \leq |x| \leq 1$$

μ Law Equation



Overview of the μ -law Compression Graph : (EXPLAIN THE GRAPH!)

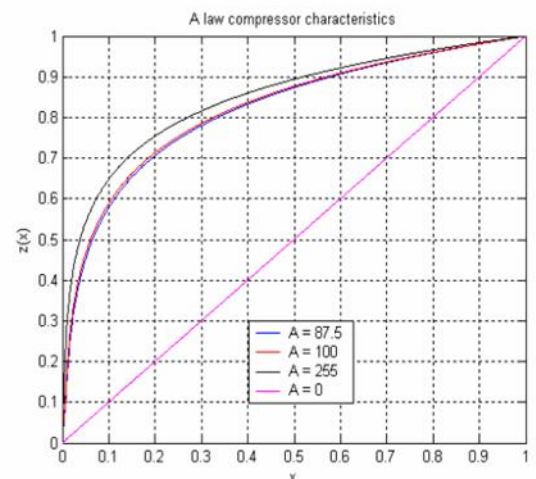
- ✓ X-axis: Represents the input signal amplitude (x), ranging from 0 to 1.
- ✓ Y-axis: Represents the output signal amplitude ($z(x)$), also ranging from 0 to 1.
- ✓ Expansion for Low Inputs: Low input values like 0.1 are expanded, resulting in a higher output value. This makes quiet sounds more pronounced.
- ✓ Compression for High Inputs: High input values like 0.9 are compressed, resulting in a lower output value. This reduces the risk of distortion and keeps loud sounds at a manageable level.

2. A-law Companding

- **Used In:** Mainly in Europe and other international systems.
- **How It Works:**
 - **Compression:** Similar to μ -law but uses a different formula to compress the signal. A-law is also logarithmic but provides a more uniform response.
 - **Formula:** The A-law compression formula is:

$$F(x) = \text{sgn}(x) \begin{cases} \frac{A|x|}{1 + \ln(A)}, & |x| < \frac{1}{A}, \\ \frac{1 + \ln(A|x|)}{1 + \ln(A)}, & \frac{1}{A} \leq |x| \leq 1, \end{cases}$$

where A is typically set to 87.6.



4. Encoder

The digitization of analog signal is done by the encoder. It designates each quantized level by a binary code. The sampling done here is the sample-and-hold process. These three sections will act as an analog to the digital converter. Encoding minimizes the bandwidth used to represent the digital data into line code there are different encoding technique.

Digital Waveforms <Line codes>

<Q> What are the desirable properties of digital waveforms?

To transmit a bit sequence 10011011, Draw the resulting waveforms using- Unipolar NRZ ; polar NRZ ; Unipolar RZ ; Bipolar RZ ; Manchester (split phase). 8 Marks

Definition: Digital Waveforms / Line codes:

↳ A digital waveform/Line code is a pattern of voltage

⊗ current used to represent digital data.

* properties of Digital Waveform:

↳ Digital waveform/signal is used to represent data as a sequence of discrete values.

↳ At any given instant it takes at most, one of a finite number of values. The duration of each level/bit same same.

↳ It can have two possible values. Therefore it is called binary signal ⊗ logic signal. one value is near a reference value (usually ground/0 volts) and the other value is near the supply voltage ($V_{CC}=5V$).

The following FIVE Basic line codes / (digital waveforms) are used to represent the digital data. They are .

<i> UNIPOLAR NRZ (Non Return to zero)

<ii> polar NRZ

<iii> Unipolar RZ (Return to zero)

<iv> Bipolar RZ

<v> Manchester (split phase)

Let us consider a bit sequence: 1 0 0 1 1 0 1 1

1. Unipolar NRZ :-

If $x(t)$ be the digital format waveform. Then Unipolar NRZ format is defined as follows:

Symbol '1': $x(t) = +A ; 0 \leq t \leq T_b$

Symbol '0': $x(t) = 0 ; 0 \leq t \leq T_b$

2. POLAR NRZ :-

Symbol '1': $x(t) = +A ; 0 \leq t \leq T_b$

Symbol '0': $x(t) = -A ; 0 \leq t \leq T_b$

3. UNIPOLAR RZ :-

Symbol '1': $x(t) = +A ; 0 \leq t \leq \frac{T_b}{2}$
 $0 ; \frac{T_b}{2} < t \leq T_b$

Symbol '0': $x(t) = 0 ; 0 \leq t \leq T_b$

4. BIPOLAR RZ :-

Symbol '1': $x(t) = +A ; 0 \leq t \leq \frac{T_b}{2}$
 $0 ; \frac{T_b}{2} < t \leq T_b$

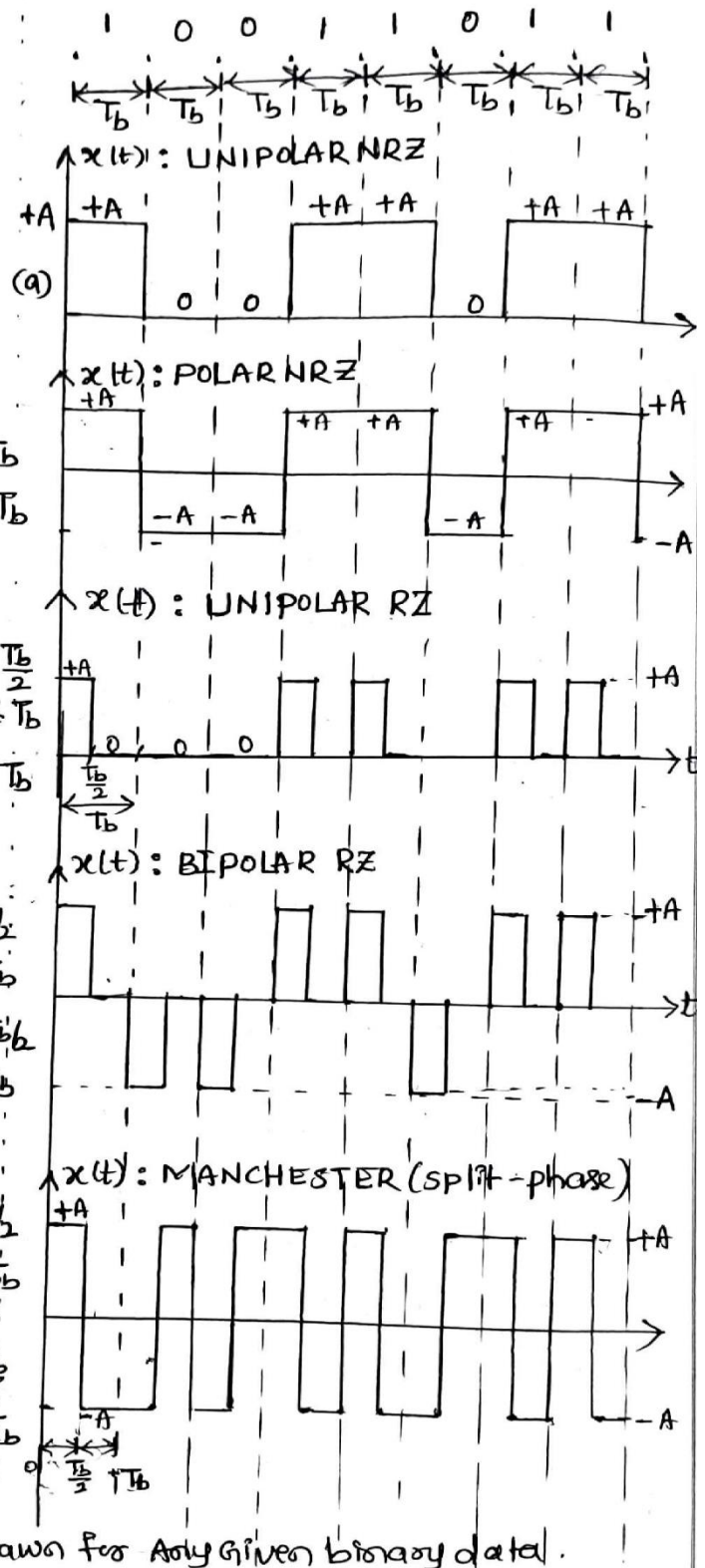
Symbol '0': $x(t) = -A ; 0 \leq t \leq \frac{T_b}{2}$
 $0 ; \frac{T_b}{2} < t \leq T_b$

5. Manchester (split phase) :-

Symbol '1': $x(t) = \begin{cases} +A & 0 \leq t \leq \frac{T_b}{2} \\ -A & \frac{T_b}{2} < t \leq T_b \end{cases}$

Symbol '0': $x(t) = \begin{cases} -A & 0 \leq t \leq \frac{T_b}{2} \\ +A & \frac{T_b}{2} < t \leq T_b \end{cases}$

Note: Using above time coding techniques, digital wlf's can be drawn for any given binary data.



5. Regenerative Repeater

The output of the channel has one regenerative repeater circuit to compensate the signal loss and reconstruct the signal. It also increases the strength of the signal.

- ① This is one of the most important features of PCM
- ② It provides the ability to control the effects of distortion and noise
- ③ It is done by a chain of **Regenerative Repeaters** located at sufficiently close spacings along the transmission route

Notes

While it is not possible to compensate for the quantization process due to the lost values, the regeneration process can overcome the effects of:

- Attenuation
- Distortion
- Random noise

This is done through:

- Detection
- Regeneration

6. Decoder

The decoder circuit decodes the pulse coded waveform to reproduce the original signal. This circuit acts as the **demodulator**.

- This process is done after regenerating the received pulses one last time
- Decoding results a **Quantized PAM** signal

7. Reconstruction Filter

After the digital-to-analog conversion is done by the regenerative circuit and the decoder, a low pass filter is employed, called as the reconstruction filter to get back the original signal.

Definition

It is the process of passing the decoder's output through a LPF with cutoff frequency equal to the message bandwidth.

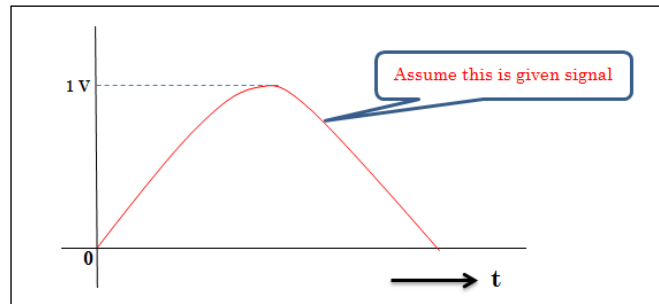
Note: For an error-free transmission path, the recovered signal is similar to the original signal with the exception of distortion introduced by the quantization process.

Hence, the Pulse Code Modulator circuit digitizes the analog signal given, codes it, and samples it. It then transmits in an analog form. This whole process is repeated in a reverse pattern to obtain the original signal.

Uniform Quantization Example :

Q. A 2-bit PCM modulator is used with a 0-1 V signal. What is the binary digital value that will occur for the following inputs: 0.4 V, 0.78 V. What is the quantization error for these two samples?

Sol: $n = \text{number of bits} = 2$
 $q = \text{No. of quantization levels} = 2^n = 4$



STEPS IN QUANTIZATION

1. Total voltage range is divided into q equal intervals of step size S

$$S = \frac{V_H - V_L}{q} = \frac{V_H - V_L}{2^n}$$

Where:

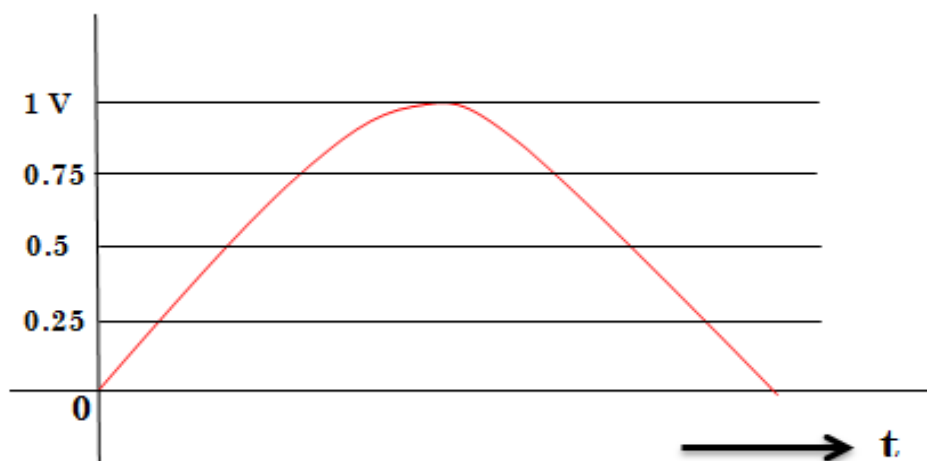
V_H = Max. voltage value

V_L = Min. voltage value

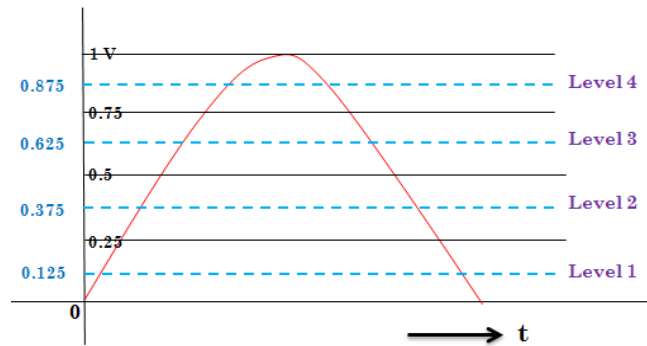
2. Draw mid lines representing quantization levels
3. Assign binary codes (pre-defined) to each quantization level
4. Calculate quantization error

Step 1 :

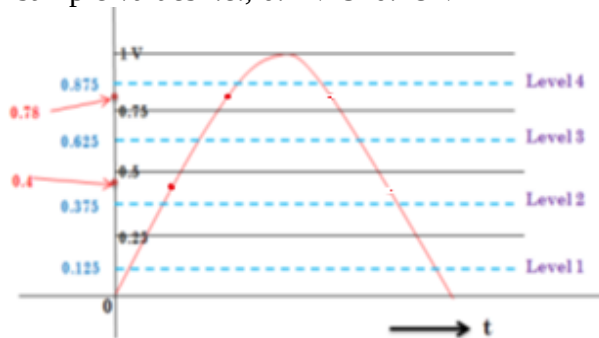
$$\text{Step size } S = \frac{V_H - V_L}{2^n} = \frac{1 - 0}{4} = \frac{1}{4} = 0.25$$



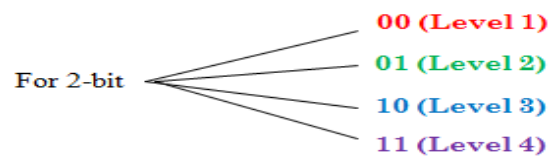
Step 2: Draw mid-lines, which represents quantization levels



Step 3: Denote given sample values i.e., 0.4 V & 0.78 V



Step 4: Represent each quantization level with predefined binary code



- **Quantizing** 0.4 V sample value: See that level 2 [0.375 v] (is near to 0.4 V. So, for sample voltage 0.4 V, 01 code is transmitted.

$$\text{Quantization error, } e = 0.4 - 0.375 = 0.025 \text{ V}$$

- **Quantizing** 0.78 V sample: Level 4 is nearest to 0.78 V. So, digital code 11 is transmitted.

$$\text{Quantization error, } e = 0.875 - 0.78 = 0.095 \text{ V}$$

Bit sequence for two samples : [11 01], we can represent it with different line coding techniques (try by your self)

Advantages of PCM.

1. Immune to channel induced noise and distortion.
2. Repeaters can be employed along the transmitting channel.
3. Encoders allow secured data transmission.
4. It ensures uniform transmission quality.

Disadvantages of PCM.

1. Pulse code modulation increases the transmission bandwidth.
2. A PCM system is somewhat more complex than another system.
3. Increases system complexity.

Time Division Multiplexing (TDM) .

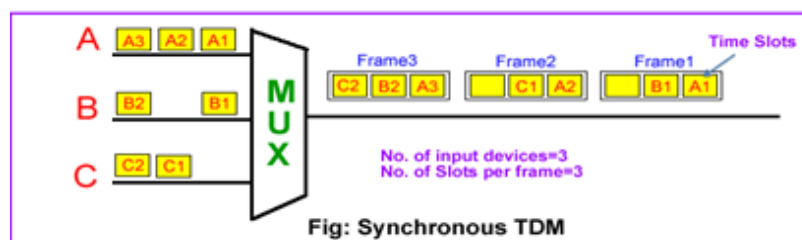
TDM is a technique used in digital modulation to transmit multiple digital signals over a single communication channel by allocating specific time slots to each signal. This allows several signals to share the same channel without interference.

Types of TDM:

- Synchronous TDM (Conventional TDM)
- Statistical TDM (Asynchronous TDM)

Parameter	Synchronous TDM	Statistical TDM
Working	Fixed time slots, even if no data is available, leading to potential bandwidth waste.	Dynamic time slots based on data availability.
No. of Slots	Fixed number of slots per frame, which can result in unused capacity.	Variable number of slots based on needs.
Buffers	Buffers data for all inputs.	Buffers data only when devices have data ready.
Data Handling	Each slot carries a specific input's data, even if it's empty, wasting bandwidth.	Slots carry data dynamically as needed.
Synchronization	All inputs start simultaneously.	No strict timing needed; flexible transmission.
Capacity	Maximum usage is fixed, leading to possible unused time.	Utilizes capacity based on actual data needs.
Data Separation	Fixed slots can discard unused data.	Only used slots are sent to the destination, necessitating the addition of ID bits to identify each device.

FIGURE: shows the different between two types :



Example: Synchronous TDM Given:

Number of input devices: 3 (A, B, C) , Number of slots per frame: 3 , **Bandwidth:** 1000 bits per second

- Assume the total time for one complete cycle (frame) of all slots is 12 ms.
1. **Total Bits per Frame:** Total Bits=Bandwidth × Frame Duration

$$\text{Total Bits} = 1000 \text{ bits/second} \times 0.012 \text{ seconds} = 12 \text{ bits}$$

2. Bits per Slot:

$$\text{Bits per Slot} = \text{Total Bits} / \text{Number of Slots}$$

$$\text{Bits per Slot} = 12 / 3 \text{ bits} = 4 \text{ bits}$$

3. Time for Each Time Slot:

$$\text{Total Time} / \text{Number of Slots} = \text{Time Slot} = 12 \text{ ms} / 3 = 4 \text{ ms}$$

This means that each of the three devices (A, B, and C) will use a time slot of 4 ms to transmit 4 bits of data in each cycle(frame).

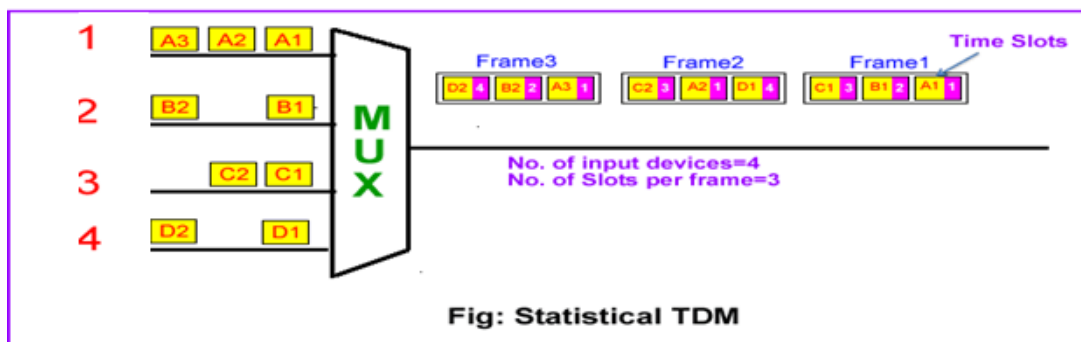


Fig: Statistical TDM

Example: Statistical TDM

Assume that Total Bandwidth: 1000 bits per second

Data Requirements with its ID :

Device A (A1): 300 bits

Device B (B1): 300 bits

Device C (C1): 400 bits

Device D (D1): 400 bits

- Total Data Total Bits to Transmit: $300 (A1) + 300 (B1) + 400 (C1) + 400 (D1) = 1400 \text{ bits}$

Transmission Time for Each Device:

$$\text{Transmission Time} = \text{Data Size} / \text{Bandwidth}$$

For Device A (A1): Transmission Time = $300 \text{ bits} / 1000 \text{ bps} = 0.3 \text{ seconds}$

For Device B (B1): Transmission Time = $300 \text{ bits} / 1000 \text{ bps} = 0.3 \text{ seconds}$

For Device C (C1): Transmission Time = $400 \text{ bits} / 1000 \text{ bps} = 0.4 \text{ seconds}$

For Device D (D1): Transmission Time = $400 \text{ bits} / 1000 \text{ bps} = 0.4 \text{ seconds}$

Total Transmission Time:

$$\text{Total Time} = 0.3 + 0.3 + 0.4 + 0.4 = 1.4 \text{ seconds}$$

Each frame can carry a total of 1000 bits. Example of Time Slot Allocation:

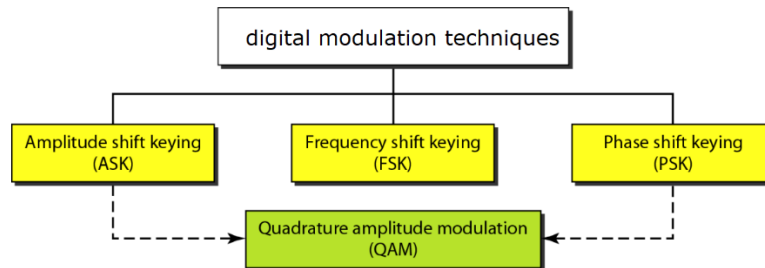
Frame 1: Device A transmits 300 bits. Device B transmits 300 bits. Device C transmits 400 bits.

Total in Frame 1 = $300 + 300 + 400 = 1000 \text{ bits}$ (fully utilized).

Frame 2: Device D transmits 400 bits. If there's no additional data, Frame 2 will be filled with 400 bits from Device D.

Digital Modulation Techniques

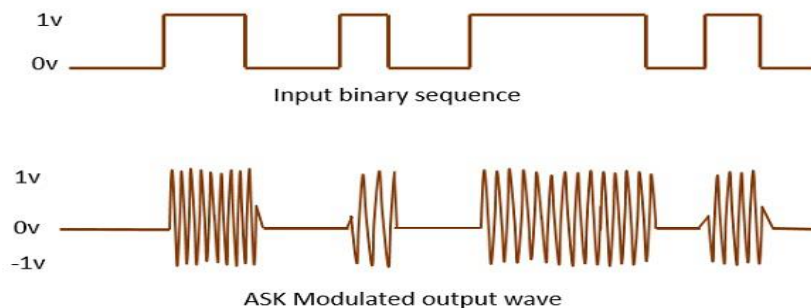
Digital Modulation provides more information capacity, high data security, quicker system availability with great quality communication. Hence, digital modulation techniques have a greater demand, for their capacity to convey larger amounts of data than analog ones. There are many types of digital modulation techniques:



Amplitude Shift Keying

1. **Amplitude Shift Keying (ASK)** is a type of Amplitude Modulation which represents the binary data in the form of variations in the amplitude of a signal.

Following is the diagram for ASK modulated waveform along with its input.



- Bandwidth of ASK :**

Bandwidth Formula The relationship between bandwidth, baud rate, and bitrate can be expressed as:

$$BW = (1+d) \cdot r$$

Where:

d: The excess bandwidth factor (a measure of how much bandwidth is used beyond the minimum required for a given data rate).

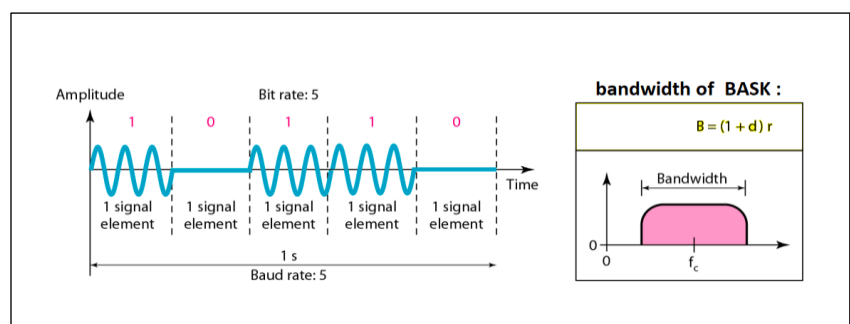
r: Baud rate Baud Rate

$$r = R/n$$

Where:

R: Bitrate (in bits per second)

n: Number of bits per symbol

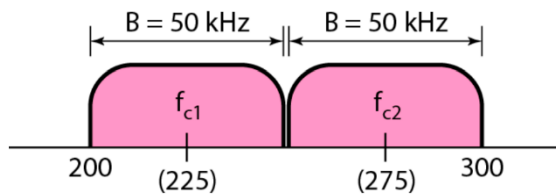


Example : We have an available bandwidth of 100 kHz which spans from 200 to 300 kHz. What are the carrier frequency and the bit rate if we modulated our data by using BASK with $d = 1$?

In Half duplex system ,The middle of the bandwidth is located at 250 kHz. This means that our carrier frequency can be at $f_c = 250$ kHz. We can use the formula for bandwidth to find the bit rate (with $d = 1$ and $n = 1$).

$$B = (1 + d) \times r = 2 \times R \times \frac{1}{n} = 2 \times R = 100 \text{ kHz} \rightarrow R = 50 \text{ kbps}$$

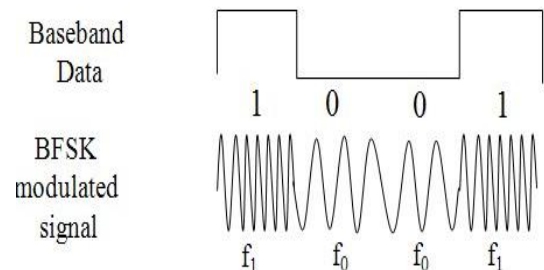
Example : In data communications, we normally use full-duplex links with communication in both directions. We need to divide the bandwidth range from 200 to 300 KHz , into two with two carrier frequencies, as shown in Figure below. The available bandwidth for each direction is now 50 kHz, which leaves us with a data rate of 25 kbps in each direction



<u>ASK Modulation:</u>	<u>ASK Demodulation :</u>
Transmitter	

2. **Frequency Shift Keying (FSK)** is the digital modulation technique in which the frequency of the carrier signal varies according to the discrete digital changes. FSK is a scheme of frequency modulation.

Following is the diagram for FSK modulated waveform along with its input.



- **Bandwidth Formula for BFSK signal can be expressed as:**

$$BW = (1+d) r + 2\Delta f$$

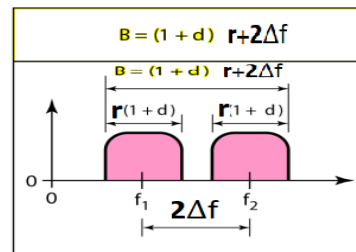
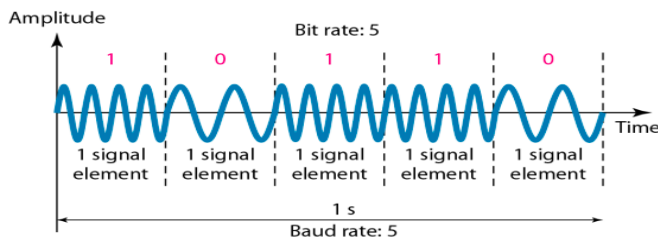
Where:

BW: Bandwidth

d: Excess bandwidth factor

r: Baud rate

- Spectrum of the BFSK :

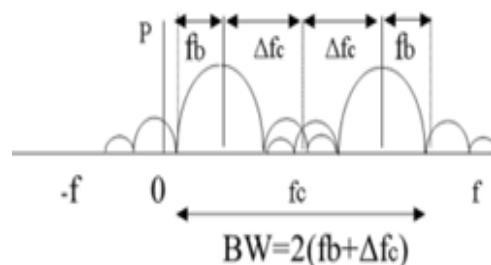


$$s(t) = \begin{cases} A \cos(2\pi f_1 t) & \text{binary 1} \\ A \cos(2\pi f_2 t) & \text{binary 0} \end{cases}$$

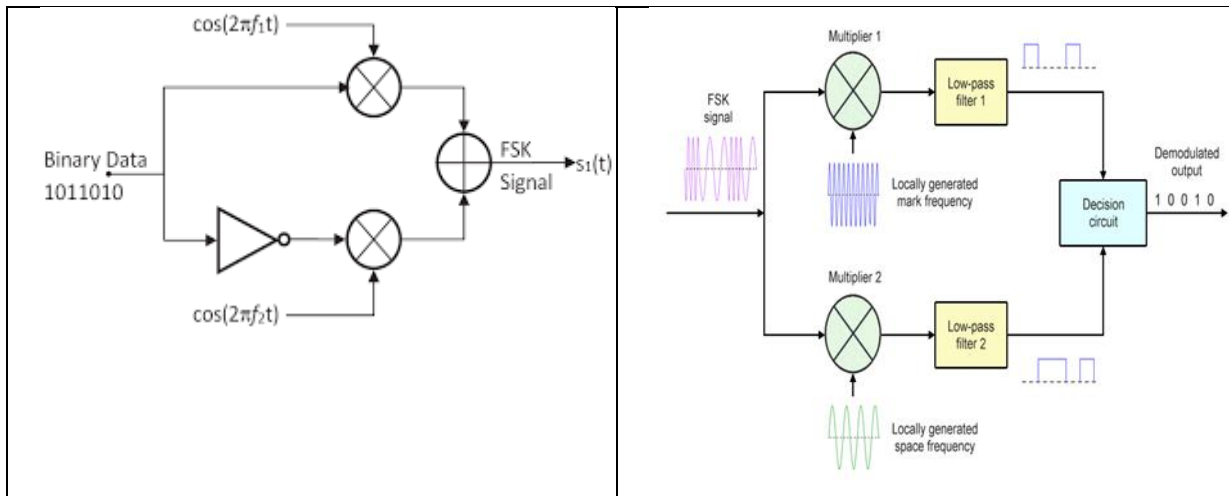
Example : We have an available bandwidth of 100 kHz which spans from 200 to 300 kHz. What should be the carrier frequency and the bit rate if we modulated our data by using BFSK with $d = 1$?

- **Solution •**

The midpoint of the band is at 250 kHz. We choose $2\Delta f$ to be 50 kHz; this means



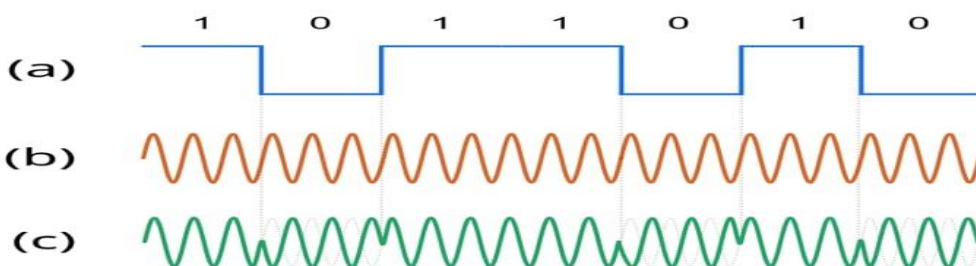
$$B = (1 + d) \times r + 2\Delta f = 100 \quad \rightarrow \quad 2r = 50 \text{ kHz} \quad r = 25 \text{ kbaud} \quad R = 25 \text{ kbps}$$



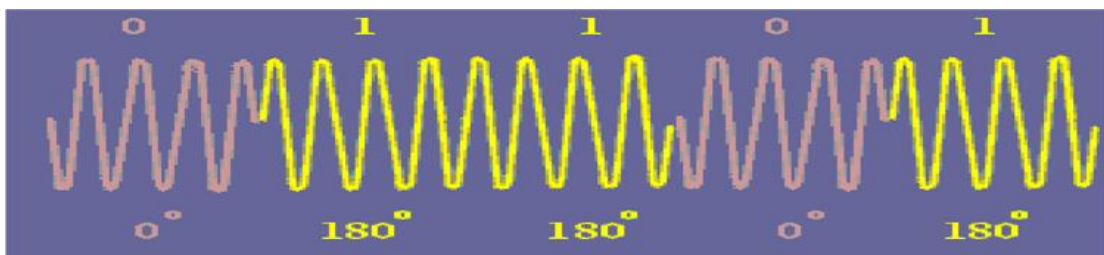
3. **Phase Shift Keying (PSK)** is the digital modulation technique in which the phase of the carrier signal is changed by varying the sine and cosine inputs at a particular time. PSK technique is widely used for wireless LANs, bio-metric along with Bluetooth communications.

Binary Phase Shift Keying (BPSK)

This is also called as **2-phase PSK** (or) **Phase Reversal Keying**. In this technique, the sine wave carrier takes two phase reversals such as 0° and 180° .



Following is the image of BPSK Modulated output wave along with its input.



- **comparison table for Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK), Phase Shift Keying (PSK):**

Feature	ASK (Amplitude Shift Keying)	FSK (Frequency Shift Keying)	PSK (Phase Shift Keying)
Modulation Type	Amplitude modulation	Frequency modulation	Phase modulation
Signal Representation	Changes amplitude of carrier signal	Changes frequency of carrier signal	Changes phase of carrier signal
Complexity	Simple	Moderate	Moderate
Bandwidth Efficiency	Low	Moderate	Moderate to High
Noise Immunity	Low	Moderate	High
Applications	Simple data transmission, RFID	Radio broadcasting, modems	Digital communication, satellite
Error Rate	Higher under noisy conditions	Moderate error rate	Lower error rate
Data Rate	Lower	Moderate	Higher

- A comparison between **Digital Modulation** and **Analog Modulation**:

Feature	Digital Modulation	Analog Modulation
Signal Type	Discrete signals (binary)	Continuous signals
Modulation Techniques	ASK, FSK, PSK, QAM	AM, FM, PM
Bandwidth Efficiency	Generally more efficient	Typically less efficient
Noise Immunity	Higher noise immunity	Lower noise immunity
Error Rate	Lower error rates with advanced coding	Higher error rates
Complexity	More complex (requires digital circuits)	Simpler (analog circuits)
Applications	Digital communication systems, data transmission	Radio broadcasting, television
Data Rate	Higher data rates	Lower data rates
Recovery of Signal	Easier to recover original signal	More difficult to recover